Calculus I

Computation problems #5

Topics:

- Optimization problems
- L'Hospital's rule
- Newton's method
- Antiderivatives

Problem 1. A rectangle has its base on the x-axis and its upper vertices on the parabola $y = 27 - x^2$. Find the maximum possible area of the rectangle.

Problem 2. Find the limit by using l'Hospital's rule

$$\lim_{x \to \pi/4} \frac{\sin x - \cos x}{x - \pi/4}.$$

Problem 3. Use Newton's method to estimate the solutions of the equation $x^2+x-1 = 0$. Start with $x_0 = -1$ for the left-hand solution and with $x_0 = 1$ for the solution on the right. Then, in each case, find x_2.

Problem 4. Explain why the following four statements ask for the same information:

(i) Find the roots of $f(x) = x^3 - 3x - 1$.

(ii) Find the x-coordinates of the intersections of the curve $y = x^3$ with the line $y = 3x + 1$.

(iii) Find the x-coordinates of the points where the curve $y = x^3 - 3x$ crosses the horizontal line $y = 1$.

(iv) Find the values of x where the derivative of $g(x) = (1/4)x^4 - (3/2)x^2 - x - 5$ equals zero.

In Problems 5–6, find an antiderivative for each function. Do as many as you can mentally. Check your answers by differentiation.

Problem 5.

a) $\dfrac{3}{2}\sqrt{x}$

b) $\dfrac{1}{2\sqrt{x}}$

c) $\sqrt{x} + \dfrac{1}{\sqrt{x}}$

Problem 6.

a) $\csc x \cot x$

b) $-\csc 5x \cot 5x$

c) $-\pi \csc \dfrac{\pi x}{2} \cot \dfrac{\pi x}{2}$

Problem 7. Find the most general antiderivative or indefinite integral. Check your answer by differentiation.

$$\int \frac{1 + \cos 4t}{2} dt$$

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Answer to selected problems

1. 108 sq units
2. $\sqrt{2}$
3. $-5/3$ and $13/21$